Topic Title: Comparison and Implementation of Data Compression Methods for SoC-based Radar Systems

Student Name: Jordan Mohindra **Student ID:** z5418455

A. **Problem statement**

In automotive radar, high-resolution cascaded MIMO sensors produce more raw data than a vehicle's network and central processor can comfortably absorb. Compressing at the sensor saves link bandwidth, but the compressor must be lossless (anything it discards is lost before target detection), keep up with the sensor, and be small enough for sensor hardware. Radar compression methods are rarely compared on the same real multi-transmitter data, and the lossless baseline was evaluated on a single-transmitter radar with no decoder.

B. **Objective**

Compare compression methods side by side on real, public, multi-transmitter radar data; select the best lossless methods; and implement their encoders and decoders on an FPGA, verified and measured on a board.

C. **My solution**

Eight methods were run in MATLAB on 50 ColoRadar frames (Texas Instruments cascade, 12 transmitters, 16 receivers) through one chain: range FFT, fixed 16-bit (FX16) scaling, compression, Doppler FFT, non-coherent integration and peak detection, against a double-precision reference. The two lossless predictive coders, DRHE and a second-order slow-time LPC, predict from the same transmitter (12 ramps back) and share an S_4/Huffman coder whose dictionary was re-derived from five frames of their prediction errors. Both, with matching decoders and a range-FFT front end, were written in C++ for Vitis HLS, built with Vivado into a DMA test system on an AMD KCU116 board at 100 MHz, and verified bit-exact from MATLAB through C-simulation and RTL co-simulation to the board.

D. **Contributions** (at most one per line, most important first)

Bit-exact lossless radar compression on an FPGA: 4.22× (DRHE), 4.40× (LPC)
Same-transmitter prediction lag for time-division MIMO radar
Comparison of eight compression methods on real cascade radar data
Breakdown of the compression ratio; Huffman dictionaries matched to the errors
Encoders, decoders and range FFT measured on the FPGA board
Reconstruction of, and comparison with, the baseline design

E. **Suggestions for future work**

Fixed-point DRHE datapath to cut logic and DSP use
Streaming (backward-adaptive) LPC without a frame buffer
All sixteen receive channels and power measurement

While I may have benefited from discussion with other people, I certify that this report is entirely my own work, except where appropriately documented acknowledgements are included.

Signature: _____ Date: _____/_____/_____

Pointers

List relevant page numbers in the column on the left. Be precise and selective: Don't list all pages of your report!

12	Problem Statement
14	Objective

Theory (up to 5 most relevant ideas)

20	Time-division MIMO radar and the data cube
21	Lossless compression and entropy coding
24	Fixed-point (FX16) quantisation
25	FPGAs and high-level synthesis
27	The baseline: Kiem's DRHE design

Method of solution (up to 5 most relevant points)

31	Eight candidate methods
32	Shared S_4/Huffman entropy coder
35	Prediction on a TDM-MIMO radar
46	Verification ladder, MATLAB to board
49	Hardware implementation

Contributions (most important first)

67	Bit-exact lossless radar compression on an FPGA
35, 63	Same-transmitter prediction lag for TDM-MIMO radar
60	Comparison of eight compression methods
64, 65	Breakdown of the ratio; matched Huffman dictionaries
68, 71	Encoders, decoders and range FFT measured on the board
74, 84	Reconstruction of, and comparison with, the baseline

My work

38–42, 49–57	System block diagrams/algorithms/equations solved
45–46	Description of assessment criteria used
44, 46, 57	Description of procedure (e.g. for experiments)

Results

72, 60	Succinct presentation of results
64, 74, 93	Analysis
91	Significance of results

Conclusion

97	Statement of whether the outcomes met the objectives
99	Suggestions for future research

Literature: (up to 5 most important references)

27–30, 74–90	[11] L. Kiem, master's thesis, 2025 (the baseline design)
23, 14	[12] A. Kramer *et al.*, ColoRadar dataset, 2022
38, 31, 26	[26] L. Li, MSc thesis, 2014 (origin of DRHE)
87, 42, 28	[27] G. Meucci and F. Mancuso, LPC-Huffman, 2022
42, 22	[15] D. A. Huffman, minimum-redundancy codes, 1952